

ASSIGNMENT #04

Course: CHE 341: CHEMICAL PROCESS CONTROL
Term: Spring 2026
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Due Date: Friday, March 20th TA's Mailbox

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

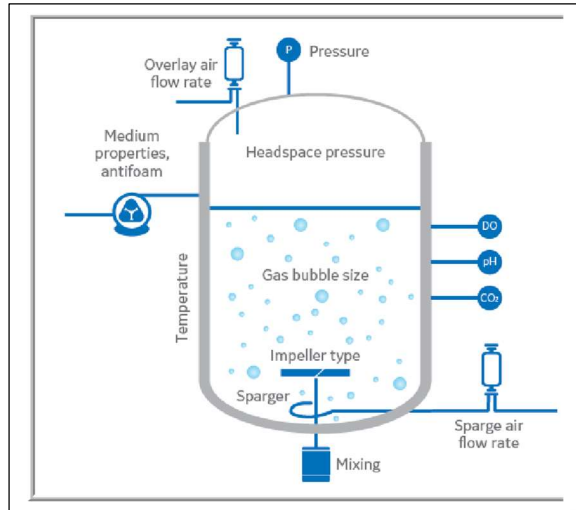
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 - Include this cover page with your name in the report.
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Chemical Process Control

Problem A: Dissolved Oxygen (DO) Process Control

Air at atmospheric pressure is injected in the bottom of a bio-reactor as shown in the schematic.



The sparger creates bubbles in the reactor while the mixer reduces their size and facilitates the transport of oxygen from the bubbles to the liquid phase (broth) which is maintained at constant temperature equal to 35 °C. The mass transfer rate of oxygen from the bubbles to the broth is given by the Newton's law.

Oxygen is consumed in the liquid phase by the cells as they react with the substrate. The rate of consumption of oxygen is given by the equation: $R_{O_2} = K x_{O_2}$ where x is the cell concentration which is assumed to be constant. The following data are available for this biological process.

Parameter	Symbol	Units	Constant Value or Formula
Reactor volume	V	m ³	1.0
Cell concentration	x	kg _{cell} /m ³	0.25
Reaction rate constant	K_{O_2}	kmol/ kg(cell s)	$1.1 \cdot 10^{-4}$
Air Flow Rate	F_{air}	ft ³ /min	500.0
Initial O concentration in 2 broth	C_{O_2}	kmol/m ³	$1.1 \cdot 10^{-4}$
Mass transfer coefficient	k_L	1/s	$0.25 + 0.001(F_{air} - 500)$

The flow rate of air in the mass transfer coefficient equation is in ft³/min. The objective is to control the dissolved oxygen (DO) concentration in the broth by manipulating the air flow rate.

1. Develop a dynamic model for the concentration of oxygen in the broth. Next taking the Laplace transform of the dynamic model and performing the proper linearization and Skögestad approximations derive the FOPTD transfer function $G_{sp}(s)$ of this process and calculate the process gain K_p , lag time θ , and time delay τ .

The rate of oxygen concentration in the broth is determined by the oxygen mass transfer from the gas bubble minus the oxygen consumption by the cells.

Which is expressed as:

$$V \frac{dC_{O_2}}{dt} = k_L a V (C^* - C_{O_2}) - R_{O_2} V$$

$$V = 1 \text{ m}^3$$

$$k_L a = 0.25 + 0.001(F_{air} - 500)$$

$$R_{O_2} = K_{O_2} x$$

The equation is divided by volume and is now expressed as:

$$\frac{dC_{O_2}}{dt} = [0.25 + 0.001(F_{air} - 500)](C^* - C_{O_2}) - K_{O_2} x$$

The model is then linearized around steady state where $\frac{dC_{O_2}}{dt} = 0$ and $F_{air} = 500$.

$$0 = 0.25(C^* - C_{O_{2,s}}) - K_{O_2} x$$

$$C_{O_{2,s}} = C^* - \frac{K_{O_2} x}{0.25}$$

$$f(C_{O_2}, F_{air}) = [0.25 + 0.001(F_{air} - 500)](C^* - C_{O_2}) - K_{O_2} x$$

$$C'_{O_2} = C_{O_2} - C_{O_{2,s}}$$

$$F'_{air} = F_{air} - F_{air,s}$$

$$\left. \frac{\partial f}{\partial C_{O_2}} \right|_s = -k_L a|_s = -0.25$$

$$\left. \frac{\partial f}{\partial F_{air}} \right|_s = 0.001(C^* - C_{O_{2,s}})$$

The Laplace transformation is then rearranged into first order:

$$sC'_{O_2} = -0.25C'_{O_2}(s) + 0.001(C^* - C_{O_{2,s}})F'_{air}(s)$$

$$\frac{C'_{O_2}(s)}{F'_{air}(s)} = \frac{0.001(C^* - C_{O_{2,s}})}{s + 0.25}$$

$$G_p(s) = \frac{K_p}{\tau s + 1}$$

$$G_p(s) = \frac{\frac{0.001(C^* - C_{O_2,s})}{0.25}}{\frac{1}{0.25}s + 1}$$

Process lag can be found through:

$$\tau = \frac{1}{0.25} = 4$$

Time delay is determined by the assumption that the measurement lag will be larger than the process lag. Therefore, based on the DO sensor the time delay is equal to thirty seconds.

$$\theta = 30 \text{ seconds}$$

Henry's Law is applied to find C^* and finalize the model. From NIST[1], the Henry's Law constant for oxygen in water at 25°C is $1.3 \times 10^{-3} \text{ mol}/(\text{kg} \cdot \text{bar})$. The Van't Law equation is used for the value at 35°C .

$$k_h(T) = k_h^\circ \times e^{C(\frac{1}{T} - \frac{1}{T^\circ})}$$

$$k_h(308.15) = 1.3 \times 10^{-3} \times e^{1500(\frac{1}{308.15} - \frac{1}{298.15})} = 1.10 \times 10^{-3}$$

$$P_{O_2} = 0.21 \times 1 \text{ atm} = 0.21 \text{ atm}$$

$$C^* = H = P_{O_2} = (1.10 \times 10^{-3}) \times 0.21 = 2.31 \times 10^{-4} \text{ kmol}/\text{m}^3$$

With these values, steady-state concentration can be defined as:

$$C^* - C_{O_2,s} = \frac{K_{O_2}x}{k_L a|_s} = \frac{(1.1 \times 10^{-4})(0.25)}{0.25} = 1.1 \times 10^{-4} \text{ kmol}/\text{m}^3$$

The Process Gain can now be calculated:

$$K_p = \frac{0.001(C^* - C_{O_2,s})}{0.25} = \frac{0.001(1.1 \times 10^{-4})}{0.25} = 4.4 \times 10^{-7} \frac{\text{kmol}/\text{m}^3}{\text{ft}^3/\text{min}}$$

2. Perform a dynamic simulation until steady-state is reached. Then perturb the flow rate of air by 10% and from the response calculate the process gain, lag time and time delay.

Problem 2 was solved using MATLAB and ODEs were performed for steady state and 10% air flow change. Process gain, lag time, and time delay were also done within the same code. The code was created with the assistance of Google Gemini[2] and will be included in the appendix further below.

The values returned were:

$$\begin{aligned}K_p &= 3.66 \times 10^{-7} \text{ (kmol/m}^3\text{)/(ft}^3\text{/min)} \\ \theta &= 2.06 \text{ seconds} \\ \tau &= 39.62 \text{ seconds}\end{aligned}$$

3. Compare the values from your dynamic simulation with the values obtained by deriving the transfer function from the Laplace transform of the dynamic model and report the percent errors.

To evaluate the dynamic models, the percent error formula was utilized, which is expressed as:

$$\text{Percent Error} = \left| \frac{\text{Simulated Value} - \text{Analytical Value}}{\text{Analytical Value}} \right| \times 100$$

The formula serves as the basis for comparing the three values: Process Gain, Time Delay, and Lag time. The simulated value will be defined as the results obtained via the Matlab code and analytical represents the theoretical data derived from hand calculations.

Process Gain (K_p):

$$\text{Percent Error} = \left| \frac{3.66 \times 10^{-7} - 4.40 \times 10^{-7}}{4.40 \times 10^{-7}} \right| \times 100 = 16.82\%$$

Time Delay (θ):

$$\text{Percent Error} = \left| \frac{2.06 - 30}{30} \right| \times 100 = 93.13\%$$

Lag Time (τ):

$$\text{Percent Error} = \left| \frac{39.62 - 4}{4} \right| \times 100 = 890.5 \%$$

Assumptions and other pertinent data:

For the necessary physical properties, you may use ASPEN Plus to obtain the appropriate parameter values and you may assume that the air contains 79% Nitrogen and 21% Oxygen. You

may also assume that the broth has the properties of water. The dissolved oxygen(DO) sensor has a lag time equal to $\tau_s = 30s$ and a repeatability of $\pm 0.5\%$



References

1. Oxygen. *NIST Chemistry WebBook*. National Institute of Standards and Technology. <https://webbook.nist.gov/cgi/cbook.cgi?ID=C7782447&Mask=10#Solubility> (accessed March 19, 2026).
2. Gemini. Google. <https://gemini.google.com/app> (accessed March 20, 2026).

Appendix

% Dynamic Simulation

clear; clc; close all;

% --- 1. Define Process Parameters ---

C_star = 2.21e-4; % Saturation DO concentration (kmol/m³)

K_O2 = 1.1e-4; % Reaction rate constant (kmol/(kg_cell*s))

X = 0.25; % Cell concentration (kg_cell/m³)

tau_s = 30; % Sensor lag time (s)

% --- 2. Phase 1: Reach Initial Steady State ---

F_air1 = 500; % Initial air flow rate

tspan1 = [0 300]; % Simulate for 300 seconds

IC1 = [11e-4; 11e-4]; % Initial guesses for broth and sensor DO

% Solve ODEs

[t1, y1] = ode45(@(t,y) bioreactor_ode(t, y, F_air1, C_star, K_O2, X, tau_s), tspan1, IC1);

% --- 3. Phase 2: Apply 10% Step Change ---

F_air2 = 550; % New air flow rate (10% increase)

```
tspan2 = [300 800]; % Simulate for next 500 seconds
```

```
IC2 = y1(end, :); % Start exactly where Phase 1 ended
```

```
% Solve ODEs
```

```
[t2, y2] = ode45(@(t,y) bioreactor_ode(t, y, F_air2, C_star, K_O2, X, tau_s), tspan2, IC2);
```

```
% --- 4. Combine Results and Plot ---
```

```
t_total = [t1; t2];
```

```
y_total = [y1; y2];
```

```
sensor_DO = y_total(:, 2); % We observe the sensor reading
```

```
figure;
```

```
plot(t_total, sensor_DO, 'b-', 'LineWidth', 2);
```

```
hold on;
```

```
xline(300, 'r--', 'Step Change Applied');
```

```
xlabel('Time (seconds)');
```

```
ylabel('Measured DO Concentration (kmol/m3)');
```

```
title('10% Air Flow Increase');
```

```
grid on;
```

```
% --- 5. Calculate FOPTD Parameters (Process Reaction Curve Method) ---
```

```
y0 = y1(end, 2);    % Initial steady state output
```

```
yf = y2(end, 2);    % Final steady state output
```

```
delta_u = F_air2 - F_air1; % Change in input (50)
```

```
delta_y = yf - y0;  % Total change in output
```

```
% Process Gain (Kp)
```

```
Kp = delta_y / delta_u;
```

```
% Find max slope for tangent line (to calculate delay and lag)
```

```
dy_dt = diff(y2(:,2)) ./ diff(t2);
```

```
[max_slope, idx] = max(dy_dt);
```

```
t_inflect = t2(idx);
```

```
y_inflect = y2(idx, 2);
```

```
% Tangent line intersections
```

```
t_delay_point = (y0 - y_inflect) / max_slope + t_inflect;
```

```
t_final_point = (yf - y_inflect) / max_slope + t_inflect;
```

```
% Time Delay (theta) and Lag Time (tau)
```

```
theta = t_delay_point - 300; % Subtract the 300s we waited before stepping
```

```
tau = t_final_point - t_delay_point;
```

```

% Plot tangent line for visualization

t_tangent = [t_delay_point, t_final_point];

y_tangent = [y0, yf];

plot(t_tangent, y_tangent, 'k--', 'LineWidth', 1.5);

legend('Sensor DO', 'Step Change', 'Tangent Line', 'Location', 'best');


% --- 6. Print Results ---

fprintf('--- Simulation Results ---\n');

fprintf('Process Gain (Kp): %.4e (kmol/m^3)/(ft^3/min)\n', Kp);

fprintf('Time Delay (theta): %.2f seconds\n', theta);

fprintf('Lag Time (tau):    %.2f seconds\n', tau);

fprintf('-----\n');


% --- Local ODE Function ---

function dydt = bioreactor_ode(t, y, F_air, C_star, K_O2, X, tau_s)

    C_broth = y(1);

    C_sensor = y(2);

    % Calculate current kLa based on flow rate

    kLa = 0.25 + 0.001 * (F_air - 500);

```

```
% ODE 1: Actual DO in broth
```

```
dC_broth = kLa * (C_star - C_broth) - (K_O2 * X);
```

```
% ODE 2: Sensor DO
```

```
dC_sensor = (C_broth - C_sensor) / tau_s;
```

```
dydt = [dC_broth; dC_sensor];
```

```
end
```
